Ayşe Betül Yüce

AI Researcher and PhD Candidate

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Profile

PhD candidate and AI researcher at AILab, Otto von Guericke University Magdeburg. Research focuses on deep learning and foundation models for time-series data, with EEG as the primary application domain. Current work examines self-supervised and representation learning, transformer-based models, transferable EEG representations, and the effects of architectural and training choices across datasets and downstream tasks.

Education

February 2023 – Present
PhD in Computer Science
AILab, Otto von Guericke University Magdeburg, Germany
Research focus: foundation models and deep learning for time-series data, with EEG as the primary application domain.

2017 – 2018
Doctoral Studies in Computer Engineering
Faculty of Computer and Informatics Engineering, Istanbul Technical University, Türkiye
Completed 24 credits of doctoral coursework and passed the doctoral qualification examination. GPA: 3.50/4.00.

2013 – 2016
M.S. in Computer Engineering
Faculty of Computer and Informatics Engineering, Istanbul Technical University, Türkiye
Thesis: *A Disagreement-Based Co-Active Learning Method for Sleep-Stage Classification*. GPA: 3.13/4.00.

2008 – 2013
B.S. in Computer Engineering
Fatih University, Türkiye
Thesis: *Unmanned Aerial Vehicle Ground Station Supervisory Control and Data Monitoring GUI Design and Implementation*. GPA: 3.68/4.00.

2008 – 2013
Double Major in Electrical and Electronics Engineering
Fatih University, Türkiye
GPA: 2.75/4.00.

Professional Experience

January 2025 – Present
AI Researcher
AILab, Otto von Guericke University Magdeburg

- Developing and evaluating EEG foundation models for speech-based mental-state detection within the AI CoWorking Lab project.
- Investigating transferable representations and explainable deep-learning methods for human-AI interaction.

February – December 2024
AI Researcher
Sport Science, Otto von Guericke University Magdeburg

- Contributed to the design of a smartphone-based data-collection experiment for analysing gait in people with spinal conditions.
- Processed accelerometer and gyroscope signals, detected gait events, and extracted features for subsequent analysis.

2020, 2021, and 2022
Visiting Researcher, DAAD–TGU Research Program
AILab, Otto von Guericke University Magdeburg

- Investigated EEG source localization using CNN, LSTM, and transformer architectures in PyTorch.
- Designed a brain-computer interface experiment using Psychtoolbox.

February 2015 – February 2024
Research Assistant
Department of Computer Science, Turkish-German University

- Contributed to academic research and supported teaching in machine learning, artificial intelligence, intelligent systems, computer organization, and algorithms and data structures.

July 2013 – January 2014
Software Developer
CRYPTTECH

- Supported the development of customer relationship management software in C#.
- Contributed to the Android version of a data-loss-prevention project.

June – July 2012
Research Intern
Fatih University

- Researched Zigbee modules, local positioning, and Kinect-based 3D mapping for an unmanned aerial vehicle project.
- Designed ground-station class diagrams and parsed GPS data for position monitoring.

June – July 2011
Software Intern
Tetaş Cyclops Information Technologies

- Developed database applications using C#.

Research Projects

January 2025 – Present
Enhancing Human–AI Collaboration through Speech-Based Mental-State Detection
OVGU AILab, Magdeburg

- Developing and evaluating EEG foundation models for mental-state detection during spoken interaction with AI systems, with a focus on transferable representations and explainable deep-learning methods.

February – December 2024
SpineInteractive
OVGU AILab and Sport Science, Magdeburg

- Contributed to the design of a smartphone-based data-collection experiment for analysing gait in people with spinal conditions.
- Processed accelerometer and gyroscope signals, detected gait events, and extracted features for subsequent analysis.

March 2020 – December 2022
EEG Source Localization
OVGU AILab, Magdeburg

- Studied forward and inverse EEG modelling using LSTM, CNN, and transformer architectures implemented in PyTorch.
- Designed a brain-computer interface experiment using Psychtoolbox.

Earlier Academic Projects

February – May 2018
Hybrid Song Recommendation System
Web Mining course project, Istanbul Technical University

- Combined matrix factorization and content-based filtering on the Million Song Dataset and Echo Nest Taste Profile subset; implemented in Python.

September – December 2017
Emotion Recognition Using Multichannel EEG Signals
Statistical Pattern Analysis and Classification course project, Istanbul Technical University

- Compared handcrafted time- and frequency-domain EEG features with a CNN operating on signal spectrograms; implemented in MATLAB, Python, and TensorFlow.

December 2014 – June 2016
A Disagreement-Based Co-Active Learning Method for Sleep-Stage Classification
Master's thesis, Istanbul Technical University

- Developed a co-active learning framework using empirical-mode-decomposition and frequency-domain feature views, evaluated on the Sleep-EDF database.

June 2012 – June 2013
Unmanned Aerial Vehicle Ground-Station GUI
Bachelor's final project, Fatih University

- Designed and implemented a C#/.NET ground-station interface and parsed GPS data to monitor UAV position.

September – December 2012
Image Colorization Using Local Binary Patterns
Pattern Recognition course project, Fatih University

- Applied machine-learning methods in WEKA to predict image colour from pixel position, grayscale intensity, and local binary patterns.

Selected Publications

1. **Yüce, A. B.**, and Stober, S. "Benchmarking Positional Encoding Strategies for Transformer-Based EEG Foundation Models." *10th Graz Brain-Computer Interface Conference*, 2026. Accepted for publication. Preprint
2. **Yüce, A. B.**, Leffler, C. J., Varghese, S., Spiliopoulou, M., and Stober, S. "Temporal Feature Extractors in EEG Foundation Models: A Controlled Comparison Including a Pretrained Time-Series Model." *2nd ICML Workshop on Foundation Models for Structured Data*, Seoul, South Korea, 2026. Paper
3. **Yüce, A. B.**, and Yaslan, Y. "A Disagreement-Based Co-Active Learning Method for Sleep Stage Classification." *23rd International Conference on Systems, Signals and Image Processing (IWSSIP)*, Bratislava, Slovakia, 23–25 May 2016. IEEE Xplore

Teaching Experience

2–6 October 2023
Instructor, Introduction to C Programming
Turkish-German University Continuing Education Center

- Designed and delivered a 20-hour intensive introductory programming course.

13–17 May 2019
Guest Lecturer, Introduction to Machine Learning
International Guest Lecturer Week, FH Bielefeld, Germany

February 2015 – February 2024
Teaching Assistant
Turkish-German University

- Supported Machine Learning, Artificial Intelligence, Intelligent Systems, Computer Organization, and Algorithms and Data Structures.

April – June 2013
Student Assistant
Electronics Laboratory, Fatih University

Professional Development

24–28 February 2025
14th Winter School on "Ethics of Neuroscience and AI"
BCCN Berlin/ICCN, Berlin School of Mind and Brain, and Cluster of Excellence "Science of Intelligence"

4–5 November 2024
Second Human-Centred Artificial Intelligence Symposium
Lukasklause, Magdeburg, Germany

6–11 September 2021
International Summer School on Situation Awareness in Cognitive Technologies (ISACT 2021)
Otto von Guericke University Magdeburg, Germany

10–12 August 2018
Data Science and Machine Learning Summer School
Boğaziçi University, Türkiye

Neural Networks and Deep Learning
Coursera certificate

Technical Skills

Programming: Python, MATLAB, C++, C#, Java, and SQL

Deep Learning and Machine Learning: PyTorch, TensorFlow, Keras, and scikit-learn

Deep Learning Methods: Foundation models, transformers, self-supervised learning, representation learning, transfer learning, CNNs, and LSTMs

Signal and Time-Series Analysis: EEG and IMU data, signal preprocessing, feature extraction, and gait-event detection

Data Analysis and Visualization: NumPy, pandas, MNE-Python and Matplotlib

Languages

Turkish: Native

English: Fluent

German: B1 (TELC)

Scholarships and Awards

2020, 2021, and 2022
DAAD–TGU Research Scholarships
Research visits to AILab, Otto von Guericke University Magdeburg

2013 – 2016
TÜBİTAK Scholarship for Master's Students
Scientific and Technological Research Council of Türkiye

2008 – 2013
Full Undergraduate Scholarship
Computer Engineering, Fatih University